MAIS PAC

IPhO 03.01: Measurement, Apparatus and Accuracy · Lecture + Problems

Measurement, Apparatus and Accuracy

MAIS Physics & Astronomy Club (PAC) · IPhO lecture + problem handout

7 lecture sections · 10 problems · 1 interactive experiment · 5 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to olympiad measurements. Five vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice. The coverage checklist is the IPhO syllabus, not this handout.

Contents & learning path

1. Measurement is comparison
2. Reading instruments properly
3. Random and systematic errors
4. Uncertainty in a single measurement
5. Propagation: the three cases that cover everything
6. Precision, accuracy and significant figures
7. Problems: reason, calculate, check
8. Further practice and provenance
9. Where to learn more

1. Measurement is comparison

Every measurement is a comparison with a standard, and every comparison has a limit. A result without an uncertainty is not a measurement — it is a guess with a decimal point.

The practical questions are always the same three:

- What is the **resolution** of the instrument (the smallest change it can show)?
- What is the **uncertainty** of this particular reading (resolution, reading error, drift, calibration)?
- How does the uncertainty **propagate** into the quantity I actually want?

2. Reading instruments properly

The most common marks lost in practical exams are not about physics at all: they are about reading an instrument and trusting it to the right number of digits.

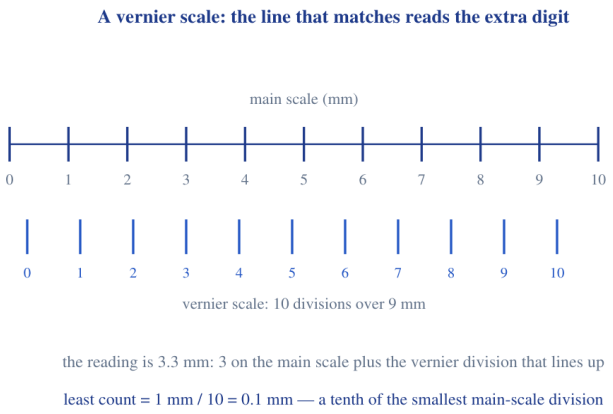


Figure 1. The vernier works by comparing two scales of slightly different pitch: exactly one division coincides, and its index gives the fraction of a main-scale division.

A **vernier calliper** has a main scale of 1 {mm} divisions and a sliding scale of 10 divisions covering 9 {mm}. Exactly one vernier division coincides with a main-scale mark, and its index gives the fraction of a millimetre. The least count is

$$\{\text{least count}\} = 1 - (9)/(10) = 0.1\ \{\text{mm}\} = (\{\text{one main division}\})/(\{\text{number of vernier divisions}\}). \tag{1}$$

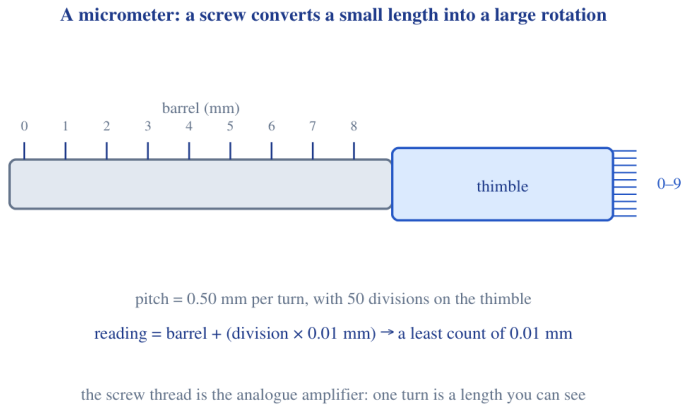


Figure 2. Both instruments beat the ruler by using a second scale rather than a finer one: the precision comes from comparison, not from sharper eyesight.

A **micrometer** uses a screw of pitch 0.50 {mm} with 50 divisions on the thimble, so one division is 0.01 {mm}. Its reading is "barrel + thimble division", and its resolution is ten times better than a vernier.

Three habits that prevent most mistakes:

- **Check the zero.** Close the jaws fully and note the reading; subtract it from every measurement. A zero error of +0.3 {mm} is invisible on a 50 {mm} object and fatal on a 1 {mm} one.
- **Avoid parallax.** Read the scale with your eye level with the pointer; a rule held away from the object introduces an error of the order of the viewing angle.
- **Use the repeatability, not the resolution, as your uncertainty.** A manual stopwatch with a 0.01 {s} display is good to about 0.2 {s} in practice.

Interactive experiment: Reading a vernier calliper — open the online handout to use it.



Figure 5. There is no "best" instrument, only an appropriate one: a micrometer used on a metre-long plank gives four meaningless digits, and a ruler used on a hair gives none.

3. Random and systematic errors

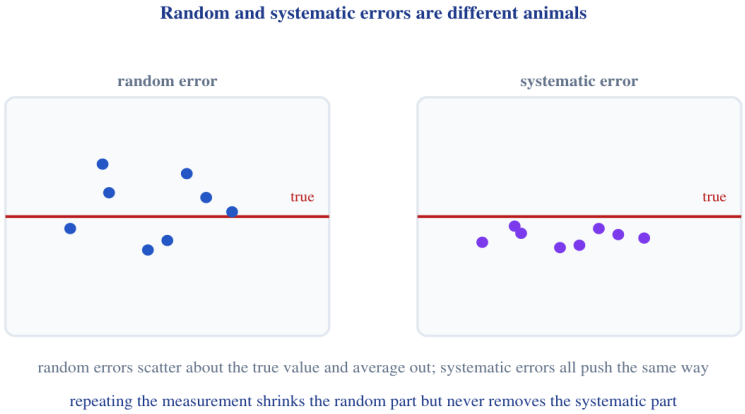


Figure 3. Random error affects precision (repeatability); systematic error affects accuracy (closeness to the truth). A calibration check, not more repeats, is what catches the systematic kind.

- **Random errors** vary in sign and size between repeats. They shift individual readings both ways, so repeating and averaging reduces their effect: the uncertainty of the mean falls as 1/√()N.
- **Systematic errors** push every reading the same way — a mis-calibrated scale, a zero error, a stopwatch started late, a systematic effect like cooling during a measurement. Repeating does not help at all; only a calibration check, a different method or a different instrument finds them.

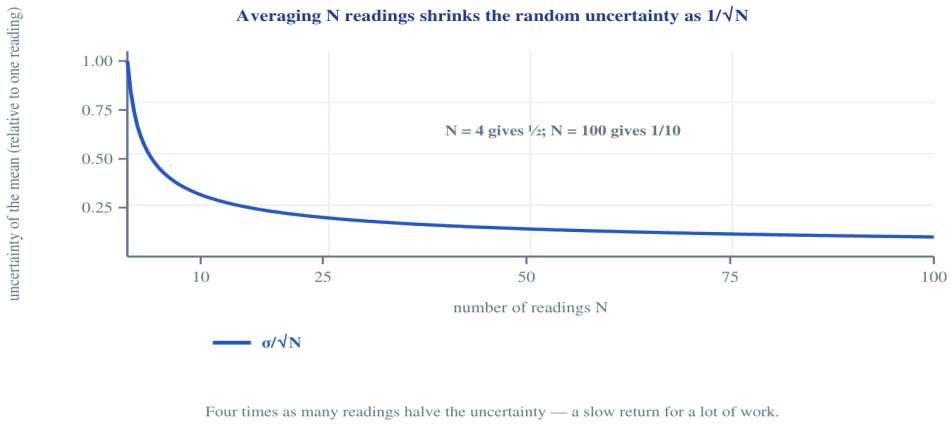


Figure. Four times as many readings halve the uncertainty — a slow return for a lot of work.

A useful way to expose a systematic error is to plot your results against a theory that **should** be a straight line through the origin. A non-zero intercept is a zero error; a slope that is consistently off is a scale error.

4. Uncertainty in a single measurement

For a single reading the uncertainty is usually taken as half the smallest division (or the whole division for a scale read between marks), plus any calibration uncertainty. Quote it to one significant figure and match the decimal places of the value:

$$L = (24.3 \pm 0.1)\ \{mm\},\ \{not\ \}\ 24.31 \pm 0.10. \tag{2}$$

5. Propagation: the three cases that cover everything

For a quantity f(x, y) with independent uncertainties Δ x, Δ y:

- **Sum or difference:** uncertainties add, Δ f = Δ x + Δ y (worst case) or √((Δ x)²+(Δ y)²) (independent random).
- **Product or quotient: fractional** uncertainties add: (Δ f)/(f) = (Δ x)/(x) + (Δ y)/(y).
- **Power:** a factor x^n multiplies the fractional uncertainty by |n|: (Δ f)/(f) = |n|(Δ x)/(x).

The most useful special case is f = x² or x³: a 1% error in a length becomes 2% in an area and 3% in a volume. That is why measuring a small dimension directly (a wire's diameter, not its cross-section) is worth the trouble.

General form (calculus): Δ f = √(((f)/(x)Δ x)² + ((f)/(y)Δ y)²).

6. Precision, accuracy and significant figures

- **Precision** = how closely repeated measurements agree (small random error).
- **Accuracy** = how close the result is to the true value (small systematic error).
- **Resolution** = the smallest increment the instrument can display.

The three are independent. A digital thermometer reading to 0.01°{C} is precise but may be 2°{C} off (inaccurate); a ruler is accurate but only precise to a millimetre; a four-digit display fed by a one-digit sensor has resolution without information.

Rules for significant figures in a final answer: keep the digits justified by the uncertainty, round the uncertainty to one significant figure, and never quote more than one extra digit in the value.

The measurement checklist. State the resolution and the uncertainty for every instrument you use → check the zero before and after → identify whether your error is random (average it away) or systematic (calibrate it away) → propagate uncertainties with the rules for sums, products and powers → quote the result with a sensible number of digits and an explicit uncertainty → sanity-check the value against a rough estimate before believing the third digit.

7. Problems: reason, calculate, check

Ten original problems in three tiers: **[1]** foundation, **[2]** multi-step olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. State assumptions, keep units, and estimate the answer before computing it. Every problem has a worked solution with a consistency check.

[1] Problem 1 — Reading a vernier. A vernier calliper has a main scale graduated in 1 {mm} and 10 vernier divisions covering 9 {mm}. Its zero reading is +0.20 {mm}. A rod gives a main-scale reading of 23 {mm} with vernier division 6 coinciding. (a) Find the least count. (b) Find the raw reading. (c) Find the corrected length with its uncertainty.

Worked solution →

[1] Problem 2 — A micrometer reading. A micrometer has a barrel graduated in 0.50 {mm} (half-millimetres) and a thimble with 50 divisions. The barrel shows 3.5 {mm} and the thimble reads 27. (a) Find the reading. (b) Find the resolution. (c) Find the percentage uncertainty in (a) and in a reading of 0.12 {mm}.

Worked solution →

[2] Problem 3 — Random or systematic? Classify each and say how you would deal with it: (a) a stopwatch started a little late every time; (b) fluctuations in the room temperature during a cooling experiment; (c) a balance that reads 0.50 {g} when empty; (d) judging the centre of a meniscus by eye; (e) the wire whose diameter you measure being slightly non-uniform.

Worked solution →

[2] Problem 4 — Propagation through a formula. The period of a pendulum is measured as T = (2.00 ± 0.02) {s}. (a) Find g using $g = 4\pi^2 L/T^2$ with L = (0.994 ± 0.002) {m}. (b) Find the percentage uncertainty in g. (c) Which measurement dominates the uncertainty, and how could you improve the experiment?

Worked solution →

[2] Problem 5 — Averaging. Ten readings of a length give a mean of 52.3 {mm} with a standard deviation of 0.12 {mm}. (a) Find the standard uncertainty of the mean. (b) Find how many readings would be needed to reduce it to 0.02 {mm}. (c) Explain why the answer to (b) is optimistic in practice.

Worked solution →

[2] Problem 6 — Calipers versus ruler. The diameter of a wire is measured with a metre rule as 2 {mm} ± 1 {mm}, and with a micrometer as 1.84 {mm} ± 0.01 {mm}. (a) Find the cross-sectional area with its uncertainty from the micrometer result. (b) Find the same from the ruler result. (c) Explain the huge difference in precision and why it matters for the resistivity.

Worked solution →

[2] Problem 7 — Significant figures. A student calculates a density as 2712.47 {kg,m^(-3)} from a mass of (54.3 ± 0.1) {g} and a volume of (20.0 ± 0.2) {cm³}. (a) Find the percentage uncertainty in the density. (b) Rewrite the result correctly. (c) Explain what is wrong with quoting all six digits.

Worked solution →

[3] Problem 8 — A systematic error hunt. A student measures the resistance of a resistor as R = V/I at five currents and gets 10.1, 10.2, 10.1, 10.3, 10.2 Ω, while the manufacturer states 9.8 Ω ± 1%. (a) Find the mean and its uncertainty. (b) Is the discrepancy statistically significant? (c) Suggest three plausible systematic errors and a test for each.

Worked solution →

[3] Problem 9 — Timing with a stopwatch. A student times 20 oscillations of a pendulum and obtains 40.2, 40.0, 40.4, 40.1, 40.3 {s}. (a) Find the mean period and its uncertainty per oscillation. (b) Explain why timing 20 oscillations rather than one is better. (c) Find how the uncertainty in the period propagates into g.

Worked solution →

[3] Problem 10 — Designing a measurement. You must determine the density of an irregularly shaped stone of mass about 30 {g} to 1%. (a) Choose the instruments and explain the choice. (b) Estimate the uncertainty achievable with each. (c) Explain which measurement limits the accuracy, and what you would do about it.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this IPhO syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IPhO syllabus (accepted 2014, amended 2015) as published by the IPhO organisation. Official problems are available through the IPhO past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.

Where to learn more

IPhO syllabus and statutes

The coverage checklist this curriculum follows: accepted 2014, amended 2015. Read it once for scope, not for depth.

IPhO past problems and solutions

The only honest test of readiness. Work recent theoretical and experimental papers under time, then compare with the official marking schemes.

BIPM: SI brochure and uncertainty guidance

The GUM is the reference behind the uncertainty rules used in 03.02 and 03.03.

Particle Data Group: mathematical tools review

A compact professional summary of means, errors, fits and confidence intervals.